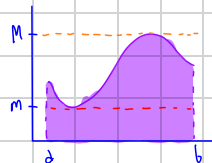


5.3

Rule (Max-min inequality)

Let $M = \max_{x \in [a,b]} f(x)$, $m = \min_{x \in [a,b]} f(x)$ 

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$73. \int_0^1 \frac{1}{1+x^2} dx : M = \frac{1}{1+0^2} = 1$$

$$m = \frac{1}{1+1^2} = \frac{1}{2}$$

$$\therefore \frac{1}{2}(1-0) \leq \int_0^1 \frac{1}{1+x^2} dx \leq 1(1-0)$$

$$75. \int_0^1 \sin(x^2) dx \leq 1(1-0) = 1$$

Recall: If $f(x) \leq g(x)$ on $[a, b]$
then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$

5.4 Fundamental Theorem of Calculus

Mean Value Theorem for Definite Integrals

→ Suppose $f(x)$ is continuous on $[a, b]$, then $\frac{1}{b-a} \int_a^b f(x) dx = f(c)$ for some $c \in [a, b]$



Proof by Theorem: let $M = \max_{x \in [a,b]} f(x)$, $m = \min_{x \in [a,b]} f(x)$, then

by Max-min Theorem, $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$

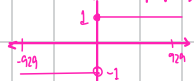
$$m \leq \frac{1}{b-a} \int_a^b f(x) dx \leq M$$

By Intermediate Value Theorem for continuous functions, there exists $c \in (a, b)$ s.t. $f(c) = \frac{\int_a^b f(x) dx}{b-a} \in [m, M]$.

e.g. suppose $f(x)$ is continuous on $[a, b]$, $\int_a^b f(x) dx = 0 \rightarrow \frac{\int_a^b f(x) dx}{b-a} = 0 \therefore$ There exists $c \in (a, b)$ s.t. $f(c) = 0$

* Continuity assumption in MVT for integrals is necessary

ex. $f(x) = \begin{cases} -1 & x < 0 \\ 1 & x \geq 0 \end{cases} \rightarrow \int_{-1}^1 f(x) dx = 0$



→ There exists $c \in [-0.99, 0.99]$ s.t. $f(c) = 0$ (But there does not exist)

(The function is discontinuous)

Fundamental Theorem Part 1

Assume $F(x)$ contin. in $[a, b]$, define $F(x) = \int_a^x f(t) dt$; $x \in [a, b]$

$$\rightarrow F(a) = 0, F(b) = \int_a^b f(t) dt$$

Then $F(x)$ is differentiable on $[a, b]$ and $F'(x) = f(x) \forall x \in [a, b]$ (F is antiderivative of f)

* Given any contin. func on $[a, b]$, it always has an antiderivative *

Proof: Consider case of $x \in (a, b)$ $\lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = \lim_{h \rightarrow 0} \frac{\int_a^{x+h} f(t) dt - \int_a^x f(t) dt}{h} = \lim_{h \rightarrow 0} \frac{\int_x^{x+h} f(t) dt}{h}$

MVT on integrals

$$\downarrow \lim_{h \rightarrow 0} f(c) \text{ for some } c \in [x, x+h]$$

$$= f(x) \text{ (} f \text{ continuous on } x \text{)}$$

$$5.4/39 \quad y = \int_0^x \sqrt{1+t^2} dt \quad \text{F.T.C. 1} \quad \frac{dy}{dx} = \sqrt{1+x^2}$$

$$5.4/41 \quad y = \int_x^{\sqrt{x}} \sin(t^2) dt = - \int_0^{\sqrt{x}} \sin(t^2) dt \quad \frac{dy}{dx} = \frac{d}{dx} - \int_0^{\sqrt{x}} \sin(t^2) dt$$

$$= - \frac{d}{dx} F(\sqrt{x}) \rightarrow - \sin(x^2)$$

$$= -F'(\sqrt{x}) \cdot (\sqrt{x})'$$

$$= -\sin(x) \cdot \frac{1}{2\sqrt{x}}$$

$$F(x) = \int_0^x \sin(t^2) dt$$

$$5.4/45 \text{ (Bruh)} \quad y = \int_{\tan x}^{\sin x} \frac{1}{\sqrt{1-t^2}} dt = \int_{\tan x}^0 \frac{1}{\sqrt{1-t^2}} dt + \int_0^{\sin x} \frac{1}{\sqrt{1-t^2}} dt$$

$$\frac{dy}{dx} = \frac{d}{dx} \left[-\int_{\tan x}^0 \frac{1}{\sqrt{1-t^2}} dt + \int_0^{\sin x} \frac{1}{\sqrt{1-t^2}} dt \right]$$

$$= -\frac{1}{\sqrt{1-\tan^2 x}} \cdot \sec^2 x + \frac{1}{\sqrt{1-\sin^2 x}} \cdot \cos x$$

$$F(x) = \int_0^x \frac{1}{\sqrt{1-t^2}} dt$$

Fundamental Theorem of Calculus Part 2

Suppose f is contin on $[a, b]$ and $F(x)$ is an antiderivative of $f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a) = F(x) \Big|_a^b$$

$$\int f(x) dx = F(x) + C$$

$$\int_a^b f(x) dx = (F(b) + C) - (F(a) + C) = F(b) - F(a)$$

Integration & Differentiation are reverse process

Proof: By FTC(I), $\left(\int_a^x f(t) dt \right)' = f(x)$

Since F is antiderivative of f , $F'(x) = f(x)$, $x \in [a, b]$

$$\rightarrow F(x) = G(x) + C \rightarrow \text{const} \quad \forall x \in [a, b]$$

$$F(b) - F(a) = (G(b) + C) - (G(a) + C)$$

$$5.4/1 \quad \int_0^2 x(x-3) dx = \int_0^2 (x^2 - 3x) dx = \left(\frac{x^3}{3} - \frac{3x^2}{2} \right) \Big|_0^2 = \frac{8}{3} - 6 = -\frac{10}{3}$$

$$26 \quad \int_0^{\pi/3} (\cos x + \sec x)^2 dx = \int_0^{\pi/3} \cos^2 x + 2 \cos x \sec x + \sec^2 x dx$$

$$\cos^2 x = \frac{\cos 2x + 1}{2} \quad = \left(\frac{\sin 2x}{4} + \frac{x}{2} + 2x + \tan x \right) \Big|_0^{\pi/3}$$

Net change theorem: Suppose $F'(x)$ exists/contin on $[a, b]$,

Corollary:

$$\int_a^b F'(x) dx = F(b) - F(a)$$

$$\int_{t_1}^{t_2} v(t) dt = s(t_2) - s(t_1)$$

$$\int_{t_1}^{t_2} s'(t) dt$$

$$\int_{t_1}^{t_2} |v(t)| dt = \text{distance travelled}$$

(Miles travelled on highway)

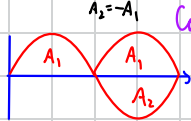
$$\text{Ex: } \int_0^{2\pi} \sin x dx = A_1 - A_2 = 0$$

$$\text{Want: } A_1 + A_2 = \int_0^{2\pi} |\sin x| dx$$

$$= 2 \int_0^{\pi} \sin x dx$$

$$= 2 (-\cos x) \Big|_0^{\pi}$$

$$= 2 (-(-1) - (-1)) = 4$$



Compared to

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

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$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

$$\int_{t_1}^{t_2} |v(t)| dt$$

5.5 Substitution method for $\int f(x) dx$

Want: $\int \odot dx$ where $\odot = f(g(x)) \cdot g'(x)$

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du \Big|_{u=g(x)}$$

Q: really true? $\rightarrow$ Let $\int f(u) du = F(u) + C$

$$F(g(x))' = f(g(x)) g'(x)$$

A: Yes!

e.g. $\int \sin^{423} x \cos x \, dx$
 Let $u = \sin x \rightarrow \int u^{423} du = \frac{u^{424}}{424} + C \Big|_{u=\sin x}$
 $= \frac{\sin^{424} x}{424} + C$

5.5/19 $\int \theta^4 \sqrt{1-\theta^2} \, d\theta$

Let $u = 1-\theta^2 \rightarrow du = -2\theta \, d\theta \rightarrow -\frac{1}{2} du = \theta \, d\theta$
 $\therefore \int \theta^4 \sqrt{1-\theta^2} \, d\theta = \int \theta^3 \cdot (\theta \, d\theta) = \int \theta^3 \cdot (-\frac{1}{2} du) = -\frac{1}{2} \int \theta^3 du$
 $= -\frac{1}{2} \cdot \frac{4}{5} u^{5/4} + C$
 $= -\frac{2}{5} u^{5/4} + C$

5.5/35 $\int \frac{1}{\theta^5} \sin(\frac{1}{\theta}) \cos(\frac{1}{\theta}) \, d\theta$

Let $u = \sin(\frac{1}{\theta}) \rightarrow du = -\frac{1}{\theta^2} \cos(\frac{1}{\theta}) \, d\theta$
 $-du = \frac{1}{\theta^2} \cos(\frac{1}{\theta}) \, d\theta$

$\int \frac{1}{\theta^5} \sin(\frac{1}{\theta}) \cos(\frac{1}{\theta}) \, d\theta = \int -u \, du \Big|_{u=\sin(\frac{1}{\theta})} = -\frac{\sin^2(\frac{1}{\theta})}{2} + C$

Suppose $f(x)$ is odd on $[-a, a] \rightarrow \int_{-a}^a f(x) \, dx = 0$

Proof Let $u = -x \rightarrow x = -u, dx = -du$

$\int f(x) \, dx = \int f(-u) (-du) = \int f(u) \, du$

Let $\int f(x) \, dx = F(x) + C \rightarrow F(x) + C = F(u) + C \therefore F(x) = F(u) = F(-x) \therefore \int f(x) \, dx$ is even in x

Suppose $f(x)$ is even on $[-a, a] \rightarrow \int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$

$\int f(x) \, dx = \int f(-u) (-du) = -\int f(u) \, du \Big|_{u=-x}$

$F(x) = -F(-x) \therefore \int f(x) \, dx$ is odd in x if $f(x)$ is even

(odd)' = even

(even)' = odd

Fact i) $\int_{-a}^a f(x) \, dx = 0$; f odd $\rightarrow \int_{-a}^a f(x) \, dx = F(x) \Big|_{-a}^a = 0$

ii) $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$; f even $\rightarrow F(a) - F(-a) = 2F(a) = \int_0^a f(t) \, dt$

Choose $F(x) = \int_0^x f(t) \, dt$
 $\rightarrow F(a) = \int_0^a f(t) \, dt$

5.6 Substitution method for $\int_a^b f(x) \, dx$

(Let $\int f(u) \, du = F(u) + C$)

Theorem $\int f(g(x)) g'(x) \, dx \stackrel{u=g(x), du=g'(x)dx}{=} \int f(u) \, du \Big|_{u=g(x)} = F(g(x)) + C$

Now, $\int_a^b f(g(x)) g'(x) \, dx = \int_a^b f(u) \, du \Big|_{u=g(x)} \stackrel{F.T.C.1}{=} F(g(x)) \Big|_a^b$

$= F(g(b)) - F(g(a)) = \int_{\text{const } g(b)}^{g(b)} f(t) \, dt - \int_{\text{const } g(a)}^{g(a)} f(t) \, dt$

$= \int_{\text{const } g(b)}^{g(b)} f(t) \, dt + \int_{g(a)}^{\text{const } g(a)} f(t) \, dt = \int_{g(a)}^{g(b)} f(t) \, dt$

5.6/2a $\int_0^1 r\sqrt{1-r^2} dr$
 $u = 1-r^2 \rightarrow du = -2r dr \rightarrow -\frac{1}{2} du = r dr$
 $\therefore \int_0^1 r\sqrt{1-r^2} dr = \int_1^0 -\frac{1}{2} \sqrt{u} du$
 $= \left(-\frac{1-r^{3/2}}{3/2} \right) \Big|_1^0 = 0 + \frac{1}{3} = \frac{1}{3}$

$$\int -\frac{\sqrt{u}}{2} du = -\frac{1}{2} \left(\frac{u^{3/2}}{3/2} \right) + C = -\frac{u^{3/2}}{3} + C$$

$$= -\frac{(1-r^2)^{3/2}}{3} + C$$

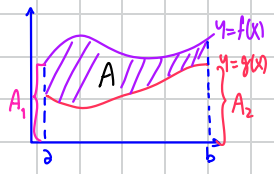
5.6/14b $\int_{-\pi/2}^{\pi/2} (2 + \tan \frac{t}{2}) \sec^2 \frac{t}{2} dt$
 Method 1 $= \int_{-\pi/2}^{\pi/2} 2 \sec^2 \frac{t}{2} dt + \int_{-\pi/2}^{\pi/2} \tan \frac{t}{2} \sec^2 \frac{t}{2} dt$
 $= \frac{2 \tan \frac{t}{2}}{\frac{1}{2}} \Big|_{-\pi/2}^{\pi/2} + 0$ (Odd function: $\sec^2 \frac{t}{2}$ even but $\tan \frac{t}{2}$ odd)
 $= 4(1 - (-1)) = 8$

Method 2 $u = 2 + \tan \frac{t}{2} \rightarrow du = \frac{1}{2} \sec^2 \frac{t}{2} dt$
 $\int_{-\pi/2}^{\pi/2} (2 + \tan \frac{t}{2}) \sec^2 \frac{t}{2} dt = \int_1^3 2u du = u^2 \Big|_1^3 = 8$
 $t = -\frac{\pi}{2} \rightarrow u = 2 + \tan(-\frac{\pi}{2}) = 2 - 1 = 1$
 $t = \frac{\pi}{2} \rightarrow u = 2 + \tan(\frac{\pi}{2}) = 2 + 1 = 3$

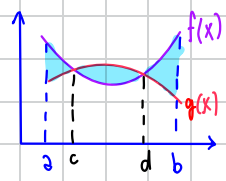
New Proof of $\int_a^b \text{odd } f(x) dx = 0$
 $\therefore \int_a^b f(x) dx \stackrel{u=x}{=} - \int_a^b f(u) du$ ($du = -dx$)
 $(\int_a^b f(u) du = \int_a^b f(x)(-dx) = \int_a^b (-f(x))(-dx) = \int_a^b f(x) dx = - \int_a^b f(x) dx)$

Area under curves

Scenario: $g(x) \leq f(x)$ on $[a, b]$ Want: $A = A_1 - A_2$
 $= \int_a^b f(x) dx - \int_a^b g(x) dx$
 $= \int_a^b (f(x) - g(x)) dx$



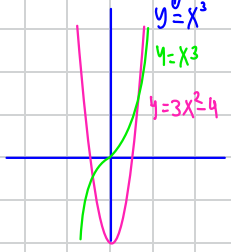
Q: What if we don't know $g(x) \leq f(x)$ on $[a, b]$



$$\therefore A = \int_a^c (f(x) - g(x)) dx + \int_c^d (g(x) - f(x)) dx + \int_d^b (f(x) - g(x)) dx$$

$$= \int_a^b |f(x) - g(x)| dx$$

5.6/60 $x^3 - y = 0$ and $3x^2 - y = 4$
 $y = x^3$
 $y = 3x^2 - 4$



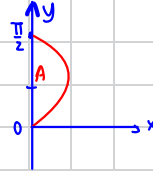
Intersections: $x^3 - 3x^2 + 4 = 0$
 $x^3 + x^2 - 4x^2 + 4 = 0$
 $x^2(x+1) - 4x(x-1) = 0$
 $(x^2 - 4x + 4)(x-1) = 0 \rightarrow x = -1, 2$

($\int_a^b f(x) dx, x=g(y) \rightarrow \int_a^b f(g(y)) g'(y) dy$)

$\int_{-1}^2 f(x) dx = \int_{-1}^2 (x^3 - (3x^2 - 4)) dx = ?$

5.6/69 $X = \int_0^{\pi/2} 3 \sin y \sqrt{\cos y} dy$
 $A = \int_0^{\pi/2} 3 \sin y \sqrt{\cos y} dy$
 $= \int_1^0 -3 \sqrt{u} du$
 $u = \cos y = 2u^{3/2} \Big|_0^1 = 2$
 $du = -\sin y dy$

$0 \leq y \leq \frac{\pi}{2}$



Chapter 6

6.1 Volumes by using slicing method

Recall



Volume = $hwl = hA$

base area



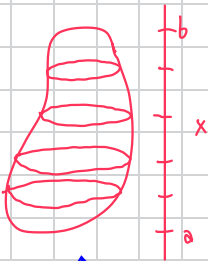
Volume = $\pi r^2 h$
 $= Ah$



General case: $A = \text{const}$, $V = hA$

Cavalieri's Principle

Same A , same $h \rightarrow$ same V

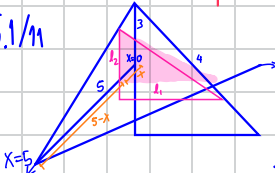


$V \approx \sum_{i=1}^n A(i) \cdot \Delta x_i$ ($A(x) = \text{area of cross section}$)

$(\sum_{i=1}^n A(i) \cdot \Delta x_i \rightarrow V \text{ as } \|P\| \rightarrow 0)$

$\therefore V = \int_a^b A(x) dx$

6.1/11

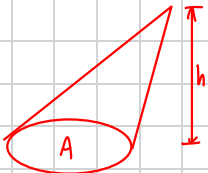


Similar to cone (Cone = $\frac{1}{3} h \cdot A \stackrel{?}{=} \frac{1}{3} \times 3 \times (\frac{1}{2} \times 4 \times 5) = 10$)

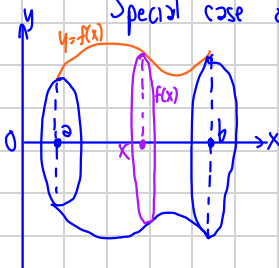
$\frac{l_1}{4} = \frac{l_2}{3} = \frac{5-x}{5}$ (Similar triangles)

$l_1 = 4(\frac{5-x}{5})$, $l_2 = 3(\frac{5-x}{5})$

$\therefore V = \int_0^5 A(x) dx = \int_0^5 \frac{1}{2} \left(\frac{20-4x}{5} \right) \left(\frac{15-3x}{5} \right) dx$
 $= \int_0^5 \frac{1}{2} \left(\frac{300-120x+12x^2}{25} \right) dx$
 $= \frac{6}{25} \int_0^5 (x^2 - 10x + 25) dx$
 $= \frac{6}{25} \left[\frac{x^3}{3} - 5x^2 + 25x \right]_0^5 = \frac{6}{25} \left(\frac{125}{3} \right) = 10$

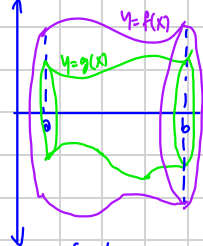


Special case of $\int_a^b A(x) dx$



rotate about X -axis

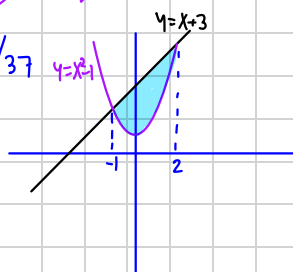
$V = \int_a^b A(x) dx = \int_a^b \pi (f(x))^2 dx$



$$V = \int_a^b \pi (f^2(x) - g^2(x)) dx$$

6.1/37

$y=x^2$



HW 5.4-5.6, 6.1

$$V = \int_{-1}^2 \pi ((x+3)^2 - (x^2)^2) dx$$